

A3 - Predicting Car Price III

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This assignment is about changing the car price problem into classification.

I use the same car dataset and cleaning steps from my previous assignments. I divide the selling price into four classes and use Multinomial Logistic Regression.

Task 1 - Classification

Import Libraries

I import the libraries that I need for this assignment.

```
In [1]: import os
import json
import time
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, confusion_matrix
```

Load the Dataset

I load the car dataset and check the first few rows.

```
In [2]: df = pd.read_csv("Cars.csv")

print("Original shape:", df.shape)
df.head()
```

Original shape: (8128, 13)

Out[2]:

	name	year	selling_price	km_driven	fuel	seller_type	transmission	
0	Maruti Swift Dzire VDI	2014	450000	145500	Diesel	Individual	Manual	
1	Skoda Rapid 1.5 TDI Ambition	2014	370000	120000	Diesel	Individual	Manual	S
2	Honda City 2017-2020 EXi	2006	158000	140000	Petrol	Individual	Manual	
3	Hyundai i20 Sportz Diesel	2010	225000	127000	Diesel	Individual	Manual	
4	Maruti Swift VXi BSIII	2007	130000	120000	Petrol	Individual	Manual	

Clean Fuel Column

I remove CNG and LPG cars so the mileage unit is more consistent.

```
In [3]: #Check fuel types
print(df['fuel'].value_counts())

#Remove CNG and LPG cars
df = df[~df['fuel'].isin(['CNG', 'LPG'])].copy()

print("\nFuel types after cleaning:")
print(df['fuel'].value_counts())
```

```
fuel
Diesel    4402
Petrol    3631
CNG         57
LPG         38
Name: count, dtype: int64
```

```
Fuel types after cleaning:
fuel
Diesel    4402
Petrol    3631
Name: count, dtype: int64
```

Clean Owner Column

I change the owner types into numbers and remove Test Drive Cars.

```
In [4]: #Change owner types into numbers
owner_map = {
    'First Owner': 1,
    'Second Owner': 2,
    'Third Owner': 3,
    'Fourth & Above Owner': 4,
    'Test Drive Car': 5
}

df['owner'] = df['owner'].map(owner_map)

#Remove Test Drive Cars
df = df[df['owner'] != 5].copy()

df['owner'].value_counts().sort_index()
```

```
Out[4]: owner
1      5238
2      2073
3       547
4       170
Name: count, dtype: int64
```

Clean Mileage, Engine and Max Power

I remove the text units and change these columns into numbers.

```
In [5]: #Clean mileage
df['mileage'] = pd.to_numeric(
    df['mileage'].str.split().str[0],
    errors='coerce'
)

#Clean engine
df['engine'] = pd.to_numeric(
    df['engine'].str.split().str[0],
    errors='coerce'
)

#Clean max power
df['max_power'] = pd.to_numeric(
    df['max_power'].str.replace('bhp', '', regex=False).str.strip(),
    errors='coerce'
)

df[['mileage', 'engine', 'max_power']].head()
```

```
Out[5]:
```

	mileage	engine	max_power
0	23.40	1248.0	74.00
1	21.14	1498.0	103.52
2	17.70	1497.0	78.00
3	23.00	1396.0	90.00
4	16.10	1298.0	88.20

Create Brand Column

I take the first word from the car name and use it as the brand.

After that I remove the full name and torque columns.

```
In [6]: #Create brand column
df['brand'] = df['name'].str.split().str[0]

#Remove name and torque columns
df = df.drop(columns=['name', 'torque'])

df.head()
```

```
Out[6]:
```

	year	selling_price	km_driven	fuel	seller_type	transmission	owner	mi
0	2014	450000	145500	Diesel	Individual	Manual	1	
1	2014	370000	120000	Diesel	Individual	Manual	2	
2	2006	158000	140000	Petrol	Individual	Manual	3	
3	2010	225000	127000	Diesel	Individual	Manual	1	
4	2007	130000	120000	Petrol	Individual	Manual	1	

Handle Missing Values

I fill the missing numerical values with the median.

```
In [7]: #Fill missing values with median
for column in ['mileage', 'engine', 'max_power', 'seats']:
    df[column] = df[column].fillna(df[column].median())

print("Cleaned shape:", df.shape)
print("\nMissing values:")
print(df.isnull().sum())
```

Cleaned shape: (8028, 12)

Missing values:

year	0
selling_price	0
km_driven	0
fuel	0
seller_type	0
transmission	0
owner	0
mileage	0
engine	0
max_power	0
seats	0
brand	0

dtype: int64

Create Four Selling Price Classes

For this assignment I do not predict the exact selling price.

I divide the selling price into four classes: 0, 1, 2 and 3. I use four groups with similar numbers of cars.

```
In [8]: #Create four price groups
df['price_class'], price_bins = pd.qcut(
    df['selling_price'],
    q=4,
    labels=False,
    retbins=True,
    duplicates='drop'
)

df['price_class'] = df['price_class'].astype(int)

print("Price class counts:")
print(df['price_class'].value_counts().sort_index())

print("\nPrice ranges:")
for c in range(4):
    print(
        f"Class {c}: "
        f"{price_bins[c]:,.0f} to {price_bins[c + 1]:,.0f}"
    )
```

```
Price class counts:
price_class
0      2050
1      2044
2      1943
3      1991
Name: count, dtype: int64
```

```
Price ranges:
Class 0: 29,999 to 260,000
Class 1: 260,000 to 450,000
Class 2: 450,000 to 680,000
Class 3: 680,000 to 10,000,000
```

The four classes have similar numbers of cars.

Class 0 is the lowest price group and Class 3 is the highest price group.

```
In [9]: #Plot price class distribution
df['price_class'].value_counts().sort_index().plot(kind='bar')
plt.xlabel('Price Class')
plt.ylabel('Number of Cars')
plt.title('Selling Price Classes')
plt.xticks(rotation=0)
plt.show()
```



Prepare X and y

I separate the input data and the price class.

I remove selling price from X because I already used it to make the price classes.

```
In [10]: #Separate X and y
X = df.drop(columns=['selling_price', 'price_class'])
y = df['price_class'].copy()

#Change text columns into numbers
X = pd.get_dummies(
    X,
    columns=['fuel', 'seller_type', 'transmission', 'brand'],
    dtype=float
)

print("X shape:", X.shape)
print("y shape:", y.shape)

X.head()
```

X shape: (8028, 46)

y shape: (8028,)

```
Out[10]:
```

	year	km_driven	owner	mileage	engine	max_power	seats	fuel_Diesel	1
0	2014	145500	1	23.40	1248.0	74.00	5.0	1.0	
1	2014	120000	2	21.14	1498.0	103.52	5.0	1.0	
2	2006	140000	3	17.70	1497.0	78.00	5.0	0.0	
3	2010	127000	1	23.00	1396.0	90.00	5.0	1.0	
4	2007	120000	1	16.10	1298.0	88.20	5.0	0.0	

5 rows × 46 columns

Split and Scale the Data

I split the data into training data and testing data.

I use 80% for training and 20% for testing. I also scale the input data before training the model.

```
In [11]: #Split the data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

#Scale the data
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("X_train_scaled shape:", X_train_scaled.shape)
print("X_test_scaled shape:", X_test_scaled.shape)
```

X_train_scaled shape: (6422, 46)
X_test_scaled shape: (1606, 46)

Add Intercept and One-Hot Target

I add one intercept column for the model.

I also change the four class labels into one-hot format.

```
In [12]: #Add intercept column
X_train_model = np.c_[
    np.ones(X_train_scaled.shape[0]),
    X_train_scaled
]

X_test_model = np.c_[
    np.ones(X_test_scaled.shape[0]),
    X_test_scaled
]

#Change y into one-hot format
k = 4
y_train_onehot = np.eye(k)[y_train.to_numpy()]

print("Training shape:", X_train_model.shape)
print("Testing shape:", X_test_model.shape)
print("One-hot target shape:", y_train_onehot.shape)
```

Training shape: (6422, 47)
Testing shape: (1606, 47)
One-hot target shape: (6422, 4)

Multinomial Logistic Regression

I use the Logistic Regression class from the class notebook and modify it for this assignment.

I add the required classification metrics and the option to use L2 penalty.

```
In [13]: class LogisticRegression:

    def __init__(
        self,
        k,
        n,
        method="batch",
        alpha=0.05,
```

```

        max_iter=500,
        penalty=None,
        l=0.0,
        random_state=42
    ):
        self.k = k
        self.n = n
        self.method = method
        self.alpha = alpha
        self.max_iter = max_iter
        self.penalty = penalty
        self.l = l
        self.random_state = random_state

    def fit(self, X, Y):
        rng = np.random.default_rng(self.random_state)

        #Start model weights
        self.W = rng.normal(
            0,
            0.01,
            size=(self.n, self.k)
        )

        self.losses = []
        start_time = time.time()

        for i in range(self.max_iter):

            if self.method == "batch":
                X_batch = X
                Y_batch = Y

            elif self.method == "minibatch":
                batch_size = int(0.3 * X.shape[0])

                idx = rng.choice(
                    X.shape[0],
                    size=batch_size,
                    replace=False
                )

                X_batch = X[idx]
                Y_batch = Y[idx]

            elif self.method == "sto":
                idx = rng.integers(0, X.shape[0])

                X_batch = X[idx:idx + 1]
                Y_batch = Y[idx:idx + 1]

            else:
                raise ValueError(
                    'method must be "batch", "minibatch" or "sto"'
                )

```

```

        loss, grad = self.gradient(
            X_batch,
            Y_batch
        )

        self.losses.append(loss)

        #Update model weights
        self.W = self.W - self.alpha * grad

        if i % 100 == 0:
            print(
                f"Loss at iteration {i}: "
                f"{loss:.6f}"
            )

        self.training_time_ = time.time() - start_time

        print(
            "Training time:",
            round(self.training_time_, 3),
            "seconds"
        )

        return self

def gradient(self, X, Y):
    m = X.shape[0]

    #Get predicted probabilities
    h = self.h_theta(X, self.W)

    #Calculate loss
    loss = -np.sum(
        Y * np.log(h + 1e-15)
    ) / m

    #Calculate error
    error = h - Y

    #Calculate gradient
    grad = self.softmax_grad(
        X,
        error
    )

    #Add L2 penalty if selected
    if self.penalty == "l2":
        W_reg = self.W.copy()

        #Do not add penalty to intercept
        W_reg[0, :] = 0.0

        loss += self.l * np.sum(
            W_reg ** 2
        )

```

```

        grad += (
            2.0 *
            self.l *
            W_reg
        )

    return loss, grad

def softmax(self, z):
    #Keep softmax values stable
    z = z - np.max(
        z,
        axis=1,
        keepdims=True
    )

    exp_scores = np.exp(z)

    return exp_scores / np.sum(
        exp_scores,
        axis=1,
        keepdims=True
    )

def softmax_grad(self, X, error):
    return (
        X.T @ error
    ) / X.shape[0]

def h_theta(self, X, W):
    return self.softmax(X @ W)

def predict(self, X):
    return np.argmax(
        self.h_theta(X, self.W),
        axis=1
    )

#Classification metrics

def support(self, y_true):
    y_true = np.asarray(y_true)

    return np.array([
        np.sum(y_true == c)
        for c in range(self.k)
    ])

def accuracy(self, y_true, y_pred):
    y_true = np.asarray(y_true)
    y_pred = np.asarray(y_pred)

    return np.mean(
        y_true == y_pred
    )

```

```

def precision(self, y_true, y_pred):
    y_true = np.asarray(y_true)
    y_pred = np.asarray(y_pred)

    scores = []

    for c in range(self.k):
        tp = np.sum(
            (y_true == c) &
            (y_pred == c)
        )

        fp = np.sum(
            (y_true != c) &
            (y_pred == c)
        )

        denominator = tp + fp

        if denominator == 0:
            score = 0.0
        else:
            score = tp / denominator

        scores.append(score)

    return np.array(scores)

def recall(self, y_true, y_pred):
    y_true = np.asarray(y_true)
    y_pred = np.asarray(y_pred)

    scores = []

    for c in range(self.k):
        tp = np.sum(
            (y_true == c) &
            (y_pred == c)
        )

        fn = np.sum(
            (y_true == c) &
            (y_pred != c)
        )

        denominator = tp + fn

        if denominator == 0:
            score = 0.0
        else:
            score = tp / denominator

        scores.append(score)

    return np.array(scores)

```

```

def f1_score(self, y_true, y_pred):
    precision = self.precision(
        y_true,
        y_pred
    )

    recall = self.recall(
        y_true,
        y_pred
    )

    denominator = precision + recall

    return np.divide(
        2 * precision * recall,
        denominator,
        out=np.zeros_like(precision),
        where=denominator != 0
    )

def macro_precision(self, y_true, y_pred):
    return np.mean(
        self.precision(
            y_true,
            y_pred
        )
    )

def macro_recall(self, y_true, y_pred):
    return np.mean(
        self.recall(
            y_true,
            y_pred
        )
    )

def macro_f1(self, y_true, y_pred):
    return np.mean(
        self.f1_score(
            y_true,
            y_pred
        )
    )

def weighted_precision(self, y_true, y_pred):
    weights = (
        self.support(y_true) /
        len(y_true)
    )

    return np.sum(
        weights *
        self.precision(
            y_true,
            y_pred
        )
    )

```

```

    )
)

def weighted_recall(self, y_true, y_pred):
    weights = (
        self.support(y_true) /
        len(y_true)
    )

    return np.sum(
        weights *
        self.recall(
            y_true,
            y_pred
        )
    )

def weighted_f1(self, y_true, y_pred):
    weights = (
        self.support(y_true) /
        len(y_true)
    )

    return np.sum(
        weights *
        self.f1_score(
            y_true,
            y_pred
        )
    )

def plot(self):
    plt.plot(
        np.arange(len(self.losses)),
        self.losses,
        label="Train Loss"
    )

    plt.title("Training Loss")
    plt.xlabel("Iteration")
    plt.ylabel("Loss")
    plt.legend()
    plt.show()

```

Support

Support means the number of real samples in each class.

For example, if there are 410 Class 0 cars in the test data, the support for Class 0 is 410.

Check My Metrics

I use a small example to compare my precision, recall and F1 functions with scikit-learn.

```
In [14]: #Small example
y_true_mock = np.array([
    0, 0, 0,
    1, 1, 1,
    2, 2, 2,
    3, 3, 3
])

y_pred_mock = np.array([
    0, 1, 0,
    1, 1, 2,
    2, 2, 3,
    3, 0, 3
])

metric_checker = LogisticRegression(
    k=4,
    n=1
)

print("My precision:", metric_checker.precision(y_true_mock, y_pred_mock))
print("My recall:", metric_checker.recall(y_true_mock, y_pred_mock))
print("My F1:", metric_checker.f1_score(y_true_mock, y_pred_mock))

print("\nscikit-learn report:")
print(
    classification_report(
        y_true_mock,
        y_pred_mock,
        labels=[0, 1, 2, 3],
        zero_division=0
    )
)
```

My precision: [0.66666667 0.66666667 0.66666667 0.66666667]

My recall: [0.66666667 0.66666667 0.66666667 0.66666667]

My F1: [0.66666667 0.66666667 0.66666667 0.66666667]

scikit-learn report:

	precision	recall	f1-score	support
0	0.67	0.67	0.67	3
1	0.67	0.67	0.67	3
2	0.67	0.67	0.67	3
3	0.67	0.67	0.67	3
accuracy			0.67	12
macro avg	0.67	0.67	0.67	12
weighted avg	0.67	0.67	0.67	12

```

In [15]: #Check that the results are the same
sk_report_mock = classification_report(
    y_true_mock,
    y_pred_mock,
    labels=[0, 1, 2, 3],
    output_dict=True,
    zero_division=0
)

for c in range(4):
    np.testing.assert_allclose(
        metric_checker.precision(
            y_true_mock,
            y_pred_mock
        )[c],
        sk_report_mock[str(c)]['precision']
    )

    np.testing.assert_allclose(
        metric_checker.recall(
            y_true_mock,
            y_pred_mock
        )[c],
        sk_report_mock[str(c)]['recall']
    )

    np.testing.assert_allclose(
        metric_checker.f1_score(
            y_true_mock,
            y_pred_mock
        )[c],
        sk_report_mock[str(c)]['f1-score']
    )

print("My metric functions match scikit-learn.")

```

My metric functions match scikit-learn.

Train Logistic Regression

First I train Logistic Regression without L2 penalty.

```

In [16]: baseline_model = LogisticRegression(
    k=4,
    n=X_train_model.shape[1],
    method='batch',
    alpha=0.05,
    max_iter=500,
    penalty=None,
    l=0.0,
    random_state=42
)

baseline_model.fit(
    X_train_model,

```

```

    y_train_onehot
)

y_pred_baseline = baseline_model.predict(
    X_test_model
)

print(
    "\nAccuracy:",
    baseline_model.accuracy(
        y_test.to_numpy(),
        y_pred_baseline
    )
)

```

```

Loss at iteration 0: 1.383841
Loss at iteration 100: 0.885335
Loss at iteration 200: 0.806630
Loss at iteration 300: 0.763882
Loss at iteration 400: 0.735441
Training time: 0.609 seconds

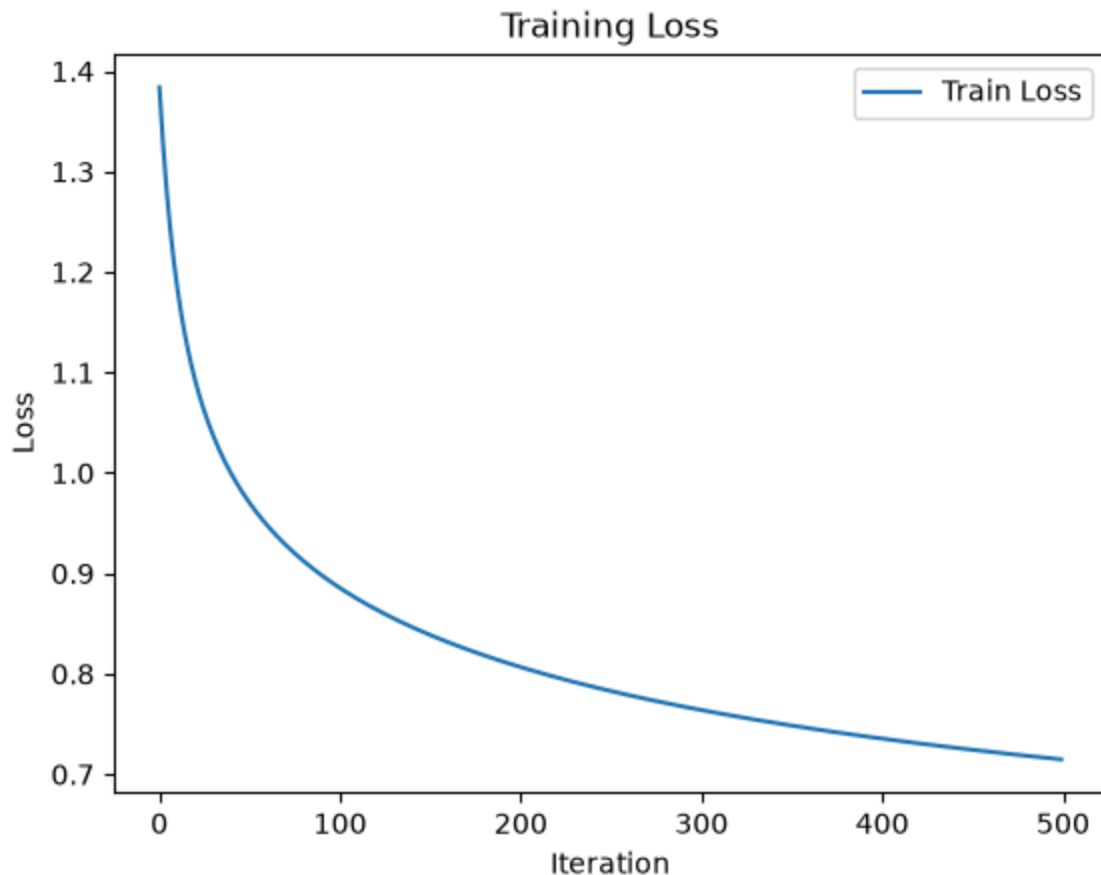
```

```
Accuracy: 0.7222914072229141
```

```

In [17]: #Plot training loss
baseline_model.plot()

```



Check Model Performance

I check accuracy, precision, recall, F1, macro average and weighted average.

```
In [18]: y_test_array = y_test.to_numpy()

precision_values = baseline_model.precision(
    y_test_array,
    y_pred_baseline
)

recall_values = baseline_model.recall(
    y_test_array,
    y_pred_baseline
)

f1_values = baseline_model.f1_score(
    y_test_array,
    y_pred_baseline
)

support_values = baseline_model.support(
    y_test_array
)

custom_report = pd.DataFrame(
    {
        'precision': precision_values,
        'recall': recall_values,
        'f1-score': f1_values,
        'support': support_values
    },
    index=[
        'Class 0',
        'Class 1',
        'Class 2',
        'Class 3'
    ]
)

custom_report
```

```
Out[18]:
```

	precision	recall	f1-score	support
Class 0	0.827103	0.863415	0.844869	410
Class 1	0.664179	0.652812	0.658446	409
Class 2	0.607046	0.575835	0.591029	389
Class 3	0.773956	0.791457	0.782609	398

```
In [19]: print(
    "Accuracy:",
    baseline_model.accuracy(
        y_test_array,
        y_pred_baseline
    )
)
```

```
)

print(
    "Macro Precision:",
    baseline_model.macro_precision(
        y_test_array,
        y_pred_baseline
    )
)

print(
    "Macro Recall:",
    baseline_model.macro_recall(
        y_test_array,
        y_pred_baseline
    )
)

print(
    "Macro F1:",
    baseline_model.macro_f1(
        y_test_array,
        y_pred_baseline
    )
)

print(
    "Weighted Precision:",
    baseline_model.weighted_precision(
        y_test_array,
        y_pred_baseline
    )
)

print(
    "Weighted Recall:",
    baseline_model.weighted_recall(
        y_test_array,
        y_pred_baseline
    )
)

print(
    "Weighted F1:",
    baseline_model.weighted_f1(
        y_test_array,
        y_pred_baseline
    )
)
)
```

Accuracy: 0.7222914072229141
 Macro Precision: 0.7180709381581021
 Macro Recall: 0.7208797830245572
 Macro F1: 0.7192382042494552
 Weighted Precision: 0.7191386816361555
 Weighted Recall: 0.7222914072229141
 Weighted F1: 0.7204790129265781

Compare with scikit-learn

I compare my results with scikit-learn classification report.

```
In [20]: print(
    classification_report(
        y_test_array,
        y_pred_baseline,
        labels=[0, 1, 2, 3],
        digits=4,
        zero_division=0
    )
)
```

	precision	recall	f1-score	support
0	0.8271	0.8634	0.8449	410
1	0.6642	0.6528	0.6584	409
2	0.6070	0.5758	0.5910	389
3	0.7740	0.7915	0.7826	398
accuracy			0.7223	1606
macro avg	0.7181	0.7209	0.7192	1606
weighted avg	0.7191	0.7223	0.7205	1606

```
In [21]: #Confusion matrix
cm = confusion_matrix(
    y_test_array,
    y_pred_baseline,
    labels=[0, 1, 2, 3]
)

plt.figure(figsize=(6, 5))
plt.imshow(cm)

plt.xticks(range(4), [0, 1, 2, 3])
plt.yticks(range(4), [0, 1, 2, 3])

plt.xlabel('Predicted Class')
plt.ylabel('True Class')
plt.title('Baseline Confusion Matrix')

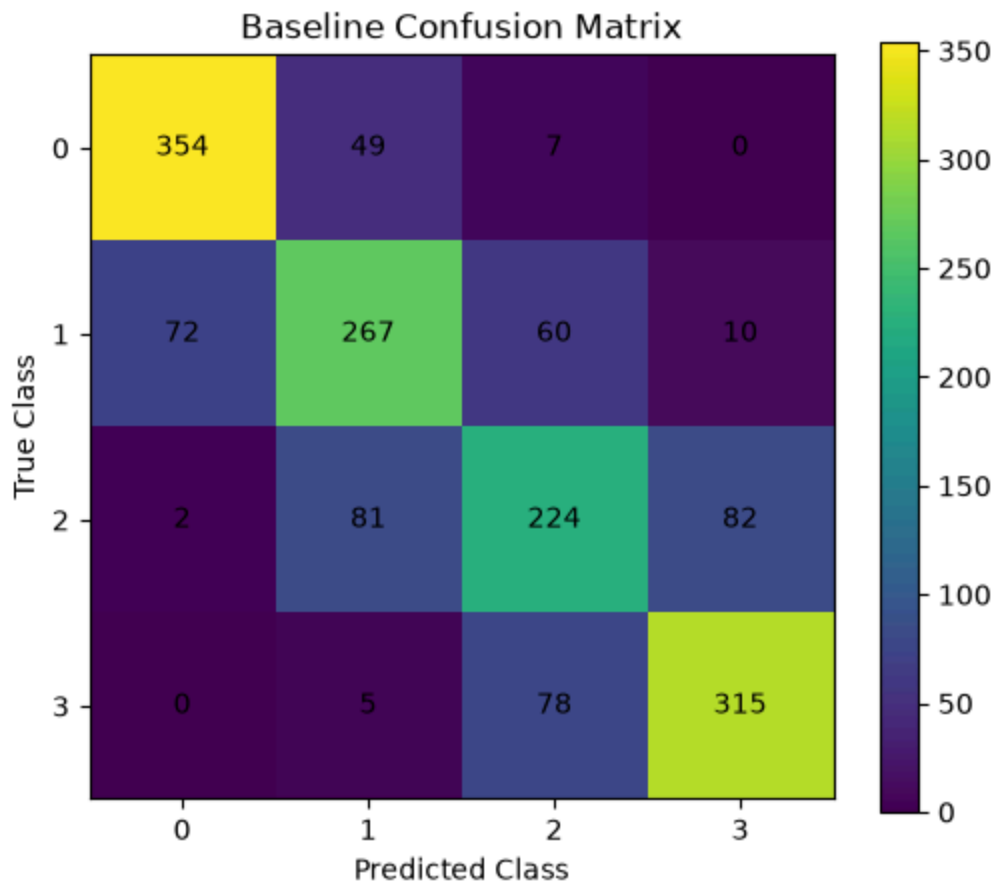
for i in range(4):
    for j in range(4):
        plt.text(
            j,
```

```

        i,
        cm[i, j],
        ha='center',
        va='center'
    )

plt.colorbar()
plt.show()

```



Task 2 - Ridge Logistic Regression

I add L2 penalty to the Logistic Regression model.

I try different lambda values and compare the results.

```

In [22]: experiment_configs = [
    {'Run': 'No penalty', 'Penalty': None, 'Lambda': 0.0},
    {'Run': 'Ridge 0.0001', 'Penalty': 'l2', 'Lambda': 0.0001},
    {'Run': 'Ridge 0.001', 'Penalty': 'l2', 'Lambda': 0.001},
    {'Run': 'Ridge 0.01', 'Penalty': 'l2', 'Lambda': 0.01},
    {'Run': 'Ridge 0.1', 'Penalty': 'l2', 'Lambda': 0.1}
]

trained_models = {}
experiment_rows = []

```

```

for config in experiment_configs:

    print("\n", config['Run'])

    model = LogisticRegression(
        k=4,
        n=X_train_model.shape[1],
        method='batch',
        alpha=0.05,
        max_iter=500,
        penalty=config['Penalty'],
        l=config['Lambda'],
        random_state=42
    )

    model.fit(
        X_train_model,
        y_train_onehot
    )

    prediction = model.predict(
        X_test_model
    )

    experiment_rows.append(
        {
            'Run': config['Run'],
            'Penalty': (
                'None'
                if config['Penalty'] is None
                else config['Penalty']
            ),
            'Lambda': config['Lambda'],
            'Accuracy': model.accuracy(
                y_test_array,
                prediction
            ),
            'Macro Precision': model.macro_precision(
                y_test_array,
                prediction
            ),
            'Macro Recall': model.macro_recall(
                y_test_array,
                prediction
            ),
            'Macro F1': model.macro_f1(
                y_test_array,
                prediction
            ),
            'Weighted F1': model.weighted_f1(
                y_test_array,
                prediction
            ),
            'Final Loss': model.losses[-1],
            'Training Time': model.training_time_
        }
    )

```

```

    )

    trained_models[config['Run']] = model

results = pd.DataFrame(
    experiment_rows
).sort_values(
    by=['Macro F1', 'Accuracy'],
    ascending=False
).reset_index(drop=True)

results

```

No penalty

```

Loss at iteration 0: 1.383841
Loss at iteration 100: 0.885335
Loss at iteration 200: 0.806630
Loss at iteration 300: 0.763882
Loss at iteration 400: 0.735441
Training time: 0.642 seconds

```

Ridge 0.0001

```

Loss at iteration 0: 1.383843
Loss at iteration 100: 0.885608
Loss at iteration 200: 0.807162
Loss at iteration 300: 0.764657
Loss at iteration 400: 0.736446
Training time: 0.65 seconds

```

Ridge 0.001

```

Loss at iteration 0: 1.383855
Loss at iteration 100: 0.888054
Loss at iteration 200: 0.811898
Loss at iteration 300: 0.771512
Loss at iteration 400: 0.745286
Training time: 0.655 seconds

```

Ridge 0.01

```

Loss at iteration 0: 1.383981
Loss at iteration 100: 0.911114
Loss at iteration 200: 0.854189
Loss at iteration 300: 0.829664
Loss at iteration 400: 0.816728
Training time: 0.655 seconds

```

Ridge 0.1

```

Loss at iteration 0: 1.385240
Loss at iteration 100: 1.053477
Loss at iteration 200: 1.048145
Loss at iteration 300: 1.047434
Loss at iteration 400: 1.047299
Training time: 0.642 seconds

```

Out[22]:

	Run	Penalty	Lambda	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F
0	No penalty	None	0.0000	0.722291	0.718071	0.720880	0.719238	0.72047
1	Ridge 0.0001	l2	0.0001	0.722291	0.718071	0.720880	0.719238	0.72047
2	Ridge 0.001	l2	0.0010	0.722291	0.718071	0.720880	0.719238	0.72047
3	Ridge 0.01	l2	0.0100	0.717933	0.713529	0.716553	0.714778	0.71599
4	Ridge 0.1	l2	0.1000	0.675592	0.668152	0.674104	0.669379	0.67055

Select the Best Model

I choose the model with the highest Macro F1 score.

```
In [23]: best_run_name = results.loc[0, 'Run']
best_model = trained_models[best_run_name]

best_prediction = best_model.predict(
    X_test_model
)

print("Best model:", best_run_name)

print(
    "Best Macro F1:",
    best_model.macro_f1(
        y_test_array,
        best_prediction
    )
)

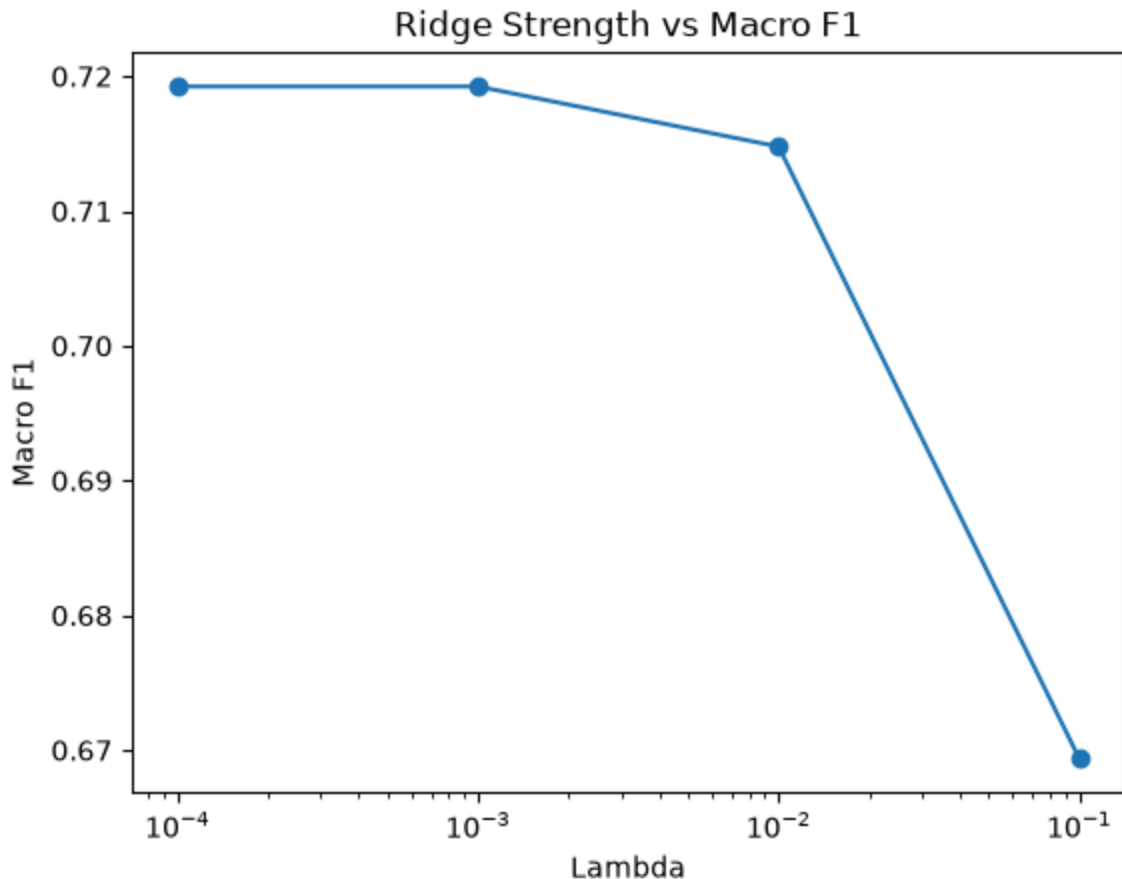
print(
    "Best Accuracy:",
    best_model.accuracy(
        y_test_array,
        best_prediction
    )
)
```

```
Best model: No penalty
Best Macro F1: 0.7192382042494552
Best Accuracy: 0.7222914072229141
```

```
In [24]: #Compare lambda and Macro F1
ridge_results = results[
    results['Penalty'] == 'l2'
].sort_values('Lambda')
```

```
plt.semilogx(
    ridge_results['Lambda'],
    ridge_results['Macro F1'],
    marker='o'
)

plt.xlabel('Lambda')
plt.ylabel('Macro F1')
plt.title('Ridge Strength vs Macro F1')
plt.show()
```



Task 2 Result

The model without penalty gave the best result with Macro F1 about 0.7192 and accuracy about 0.7223.

Ridge 0.0001 and Ridge 0.001 gave the same test score, but the stronger L2 values reduced the result. I use the model without penalty as the best model.

Task 3 - MLflow and Deployment

Now I log the experiment to the class MLflow server.

I use the experiment name `st127119-a3`. I save the best model as `st127119-a3-model`.

I do not log the dataset.

MLflow execution note: The class MLflow server was not reachable on port 443 during final testing, so the Task 3 code is included below but the MLflow logging and model registration cells could not be executed.

```
In [ ]: #Install MLflow only if it is not already installed
# %pip install "mlflow>=2.9,<3" cloudpickle
```

```
In [ ]: import os
from getpass import getpass

import mlflow
from mlflow.models import infer_signature
from mlflow.tracking import MlflowClient

#Class MLflow login
os.environ['MLFLOW_TRACKING_USERNAME'] = 'student'

if 'MLFLOW_TRACKING_PASSWORD' not in os.environ:
    os.environ['MLFLOW_TRACKING_PASSWORD'] = getpass(
        'MLflow password: '
    )

MLFLOW_TRACKING_URI = 'https://mlflow.ml.brain.cs.ait.ac.th'
EXPERIMENT_NAME = 'st127119-a3'
MODEL_NAME = 'st127119-a3-model'

mlflow.set_tracking_uri(MLFLOW_TRACKING_URI)
mlflow.set_registry_uri(MLFLOW_TRACKING_URI)
mlflow.set_experiment(EXPERIMENT_NAME)
```

Prepare Model for MLflow

I save the preprocessing and model together so the web app can use normal car information as input.

```
In [ ]: RAW_FEATURE_COLUMNS = [
    'year',
    'km_driven',
    'fuel',
    'seller_type',
    'transmission',
    'owner',
    'mileage',
    'engine',
    'max_power',
    'seats',
    'brand'
]

NUMERIC_COLUMNS = [
```

```

        'year',
        'km_driven',
        'owner',
        'mileage',
        'engine',
        'max_power',
        'seats'
    ]

    CATEGORICAL_COLUMNS = [
        'fuel',
        'seller_type',
        'transmission',
        'brand'
    ]

    numeric_medians = {
        column: float(df[column].median())
        for column in NUMERIC_COLUMNS
    }

    categorical_modes = {
        column: df[column].mode()[0]
        for column in CATEGORICAL_COLUMNS
    }

    feature_columns = X.columns.tolist()

    class CarPricePyfuncModel(mlflow.pyfunc.PythonModel):

        def __init__(
            self,
            classifier,
            scaler,
            feature_columns,
            numeric_medians,
            categorical_modes,
            price_bins
        ):
            self.classifier = classifier
            self.scaler = scaler
            self.feature_columns = feature_columns
            self.numeric_medians = numeric_medians
            self.categorical_modes = categorical_modes
            self.price_bins = np.asarray(
                price_bins,
                dtype=float
            )

        def predict(
            self,
            context,
            model_input
        ):
            frame = pd.DataFrame(

```

```

        model_input
    ).copy()

    frame = frame[
        RAW_FEATURE_COLUMNS
    ].copy()

    for column, median in self.numeric_medians.items():
        frame[column] = (
            pd.to_numeric(
                frame[column],
                errors='coerce'
            )
            .fillna(median)
        )

    for column, mode in self.categorical_modes.items():
        frame[column] = (
            frame[column]
            .fillna(mode)
            .astype(str)
        )

    encoded = pd.get_dummies(
        frame,
        columns=CATEGORICAL_COLUMNS,
        dtype=float
    )

    encoded = encoded.reindex(
        columns=self.feature_columns,
        fill_value=0.0
    )

    scaled = self.scaler.transform(
        encoded
    )

    model_X = np.c_[
        np.ones(scaled.shape[0]),
        scaled
    ]

    classes = (
        self.classifier
        .predict(model_X)
        .astype(int)
    )

    lower_bounds = np.array([
        self.price_bins[c]
        for c in classes
    ])

    upper_bounds = np.array([
        self.price_bins[c + 1]

```

```

        for c in classes
    ])

    return pd.DataFrame(
        {
            'price_class': classes,
            'lower_bound': lower_bounds,
            'upper_bound': upper_bounds
        }
    )

```

```

In [ ]: #Check the saved model format before MLflow
best_wrapper = CarPricePyfuncModel(
    classifier=best_model,
    scaler=scaler,
    feature_columns=feature_columns,
    numeric_medians=numeric_medians,
    categorical_modes=categorical_modes,
    price_bins=price_bins
)

sample_input = df[
    RAW_FEATURE_COLUMNS
].head(5).copy()

sample_output = best_wrapper.predict(
    None,
    sample_input
)

sample_output

```

Log Experiments to MLflow

I log the model settings and results for each experiment.

```

In [ ]: logged_runs = {}

result_lookup = (
    results
    .set_index('Run')
    .to_dict(orient='index')
)

sample_input = df[
    RAW_FEATURE_COLUMNS
].head(5).copy()

for run_name, model in trained_models.items():

    row = result_lookup[run_name]

    wrapper = CarPricePyfuncModel(
        classifier=model,

```

```

        scaler=scaler,
        feature_columns=feature_columns,
        numeric_medians=numeric_medians,
        categorical_modes=categorical_modes,
        price_bins=price_bins
    )

    output_example = wrapper.predict(
        None,
        sample_input
    )

    signature = infer_signature(
        sample_input,
        output_example
    )

    with mlflow.start_run(
        run_name=run_name
    ) as run:

        mlflow.log_params(
            {
                'method': model.method,
                'alpha': model.alpha,
                'max_iter': model.max_iter,
                'penalty': (
                    'none'
                    if model.penalty is None
                    else model.penalty
                ),
                'lambda': float(model.l)
            }
        )

        mlflow.log_metrics(
            {
                'accuracy': float(row['Accuracy']),
                'macro_precision': float(row['Macro Precision']),
                'macro_recall': float(row['Macro Recall']),
                'macro_f1': float(row['Macro F1']),
                'weighted_f1': float(row['Weighted F1']),
                'final_loss': float(row['Final Loss']),
                'training_time': float(row['Training Time'])
            }
        )

        #Save model only
        mlflow.pyfunc.log_model(
            artifact_path='model',
            python_model=wrapper,
            input_example=sample_input,
            signature=signature
        )

        logged_runs[run_name] = run.info.run_id

```

```
print(
    run_name,
    '->',
    run.info.run_id
)
```

Register the Best Model

I register the best model and move it to Staging.

```
In [ ]: best_run_id = logged_runs[
        best_run_name
    ]

best_model_uri = (
    f'runs://{best_run_id}/model'
)

registered_model = mlflow.register_model(
    model_uri=best_model_uri,
    name=MODEL_NAME
)

client = MlflowClient()

client.transition_model_version_stage(
    name=MODEL_NAME,
    version=registered_model.version,
    stage='Staging'
)

print('Registered model:', MODEL_NAME)
print('Version:', registered_model.version)
print('Stage: Staging')
```

Deployment Files and CI/CD

I prepare the files for the web app and GitHub.

I use a README file, an app folder, two unit tests and GitHub Actions. The tests check the model input and output shape.

GitHub Actions runs the tests when I push the code. If the tests pass, deployment can continue.